Exp:4. Handling Missing and Inappropriate Data in a Dataset

Code:

```
import numpy as np
import pandas as pd

# Creating the original dataset
df = pd.DataFrame({
    'CustomerID': [1, 2, 3, 4, 5],
    'Age-group': ['20-25', '30-35', '25-30', '20-25', '35+'],
    'Rating (1-5)': [4, 5, 6, -1, 3],
    'Hotel': ['Ibis', 'Lemon Tree', 'RedFox', 'Lemon Tree', 'Ibis'],
    'FoodPreference': ['veg', 'Non-veg', 'veg', 'veg', 'Vegetarian'],
    'Bill': [1300, 2000, 1322, 1234, 989],
    'NoOfPax': [2, 3, 2, 2, 2],
    'Estimated Salary': [40000, 59000, 30000, 1200000, 45000],
    'Age_group.1': ['20-25', '30-35', '25-30', '20-25', '35+']
})

print("ORIGINAL DATASET")
print(df)

print("\nDUPLICATE ROWS")
# (Assuming df.duplicated() goes here based on context)
```

```
df.reset_index(drop=True, inplace=True)
df.drop(['Age_group.1'], axis=1, inplace=True)

df.loc[(df['Rating (1-5)'] < 1) | (df['Rating (1-5)'] > 5), 'Rating (1-5)'] =
np.nan

df.loc[df['Bill'] < 0, 'Bill'] = np.nan
df.loc[df['Estimated Salary'] < 0, 'Estimated Salary'] = np.nan
df.loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20), 'NoOfPax'] = np.nan

print("\nUNIQUE AGE GROUPS")
print(df['Age-group'].unique())
print("\nUNIQUE HOTELS")
print(df['Hotel'].unique())
print("\nUNIQUE FOOD PREFERENCES")
print(df['FoodPreference'].unique())

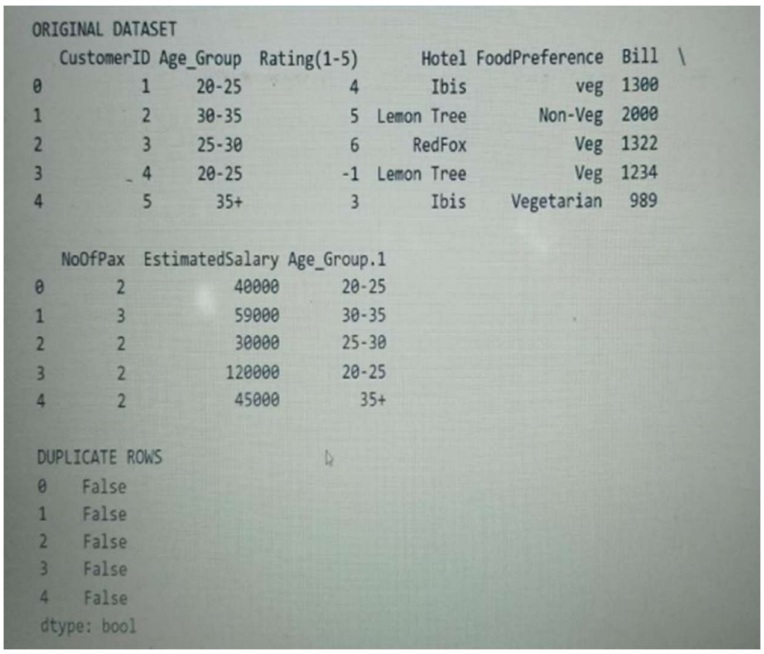
df['FoodPreference'] = df['FoodPreference'].replace(['non-veg', 'non -
veg'], 'Non-veg')

df['Estimated Salary'] = df['Estimated
Salary'].fillna(round(df['Estimated Salary'].mean()))
df['NoOfPax'] = df['NoOfPax'].fillna(round(df['NoOfPax'].median()))
df['Bill'] = df['Bill'].fillna(round(df['Bill'].mean()))
```

print("\nFINAL CLEANED DATASET")

print(df)

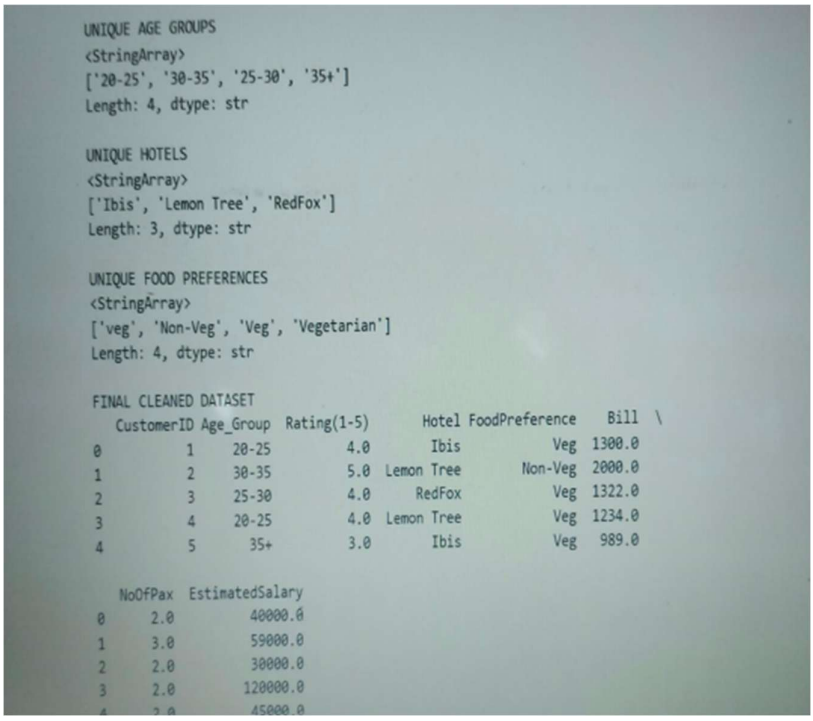
Output:



```
ORIGINAL DATASET
   CustomerID Age_Group  Rating(1-5)       Hotel FoodPreference  Bill  \
0           1     20-25           4        Ibis            veg  1300
1           2     30-35           5  Lemon Tree        Non-Veg  2000
2           3     25-30           6     RedFox            Veg  1322
3           4     20-25          -1  Lemon Tree            Veg  1234
4           5       35+           3        Ibis     Vegetarian   989

   NoOfPax  EstimatedSalary Age_Group.1
0        2            40000       20-25
1        3            59000       30-35
2        2            30000       25-30
3        2           120000       20-25
4        2            45000         35+

DUPLICATE ROWS
0    False
1    False
2    False
3    False
4    False
dtype: bool
```



```
UNIQUE AGE GROUPS
<StringArray>
['20-25', '30-35', '25-30', '35+']
Length: 4, dtype: str

UNIQUE HOTELS
<StringArray>
['Ibis', 'Lemon Tree', 'RedFox']
Length: 3, dtype: str

UNIQUE FOOD PREFERENCES
<StringArray>
['veg', 'Non-Veg', 'Veg', 'Vegetarian']
Length: 4, dtype: str

FINAL CLEANED DATASET
   CustomerID Age_Group  Rating(1-5)       Hotel FoodPreference    Bill  \
0           1     20-25          4.0        Ibis            Veg  1300.0
1           2     30-35          5.0  Lemon Tree        Non-Veg  2000.0
2           3     25-30          4.0     RedFox            Veg  1322.0
3           4     20-25          4.0  Lemon Tree            Veg  1234.0
4           5       35+          3.0        Ibis            Veg   989.0

   NoOfPax  EstimatedSalary
0      2.0          40000.0
1      3.0          59000.0
2      2.0          30000.0
3      2.0         120000.0
4      2.0          45000.0
```

Result

Handling Missing and Inappropriate Data in a Dataset analysed.